Dopaminergic SPECT estimation from structural MRI sequences to characterize Parkinson's disease.

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Abstract

Parkinson's disease is associated with the progressive death of dopaminergic neurons producing motor disorders at advanced stages. Today, there is no definitive biomarker to early detect and diagnose the disease (80% of the dopaminergic neurons are lost at first diagnosis). Recently, neuroimages have allowed to discover radiological markers associated with structural alterations from routine T1-MRI, and even measure dopamine activations in specialized SPECT sequences. Nonetheless, SPECT sequences are highly specialized, with low availability in clinical centers, and the respective analysis and quantification remain subjective and poorly explored. This work introduces a self-supervised deep representation able to capture dopaminergic levels from an MRI-to-SPECT transformation, learning an embedding representation with the capability to classify between Parkinson's and control subjects. From a retrospective study with a set of 73 Parkinson's and 73 control patients, the proposed approach achieved a notable discrimination of 79% in the precision metric.

Keywords: Parkinson's disease, Single-Photon Emission Computed Tomography (SPECT), Magnetic resonance imaging (MRI), Self-supervised learning, Non-aligned projections.

Conventions: EMBC, Franklin, **Profe**

1 Introduction

Parkinson's disease (PD) is the second most prevalent neurodegenerative disorder worldwide, impacting over ten million people. Projections promote a remarked increase in the PD population, with an estimation of around 20% at 2040 [?]. PD is associated with the progressive degeneration of dopaminergic neurons, crucial neurotransmitters that play an important role in the body's functions affecting regions such as substantia nigra pars compacta (SNc) on the striatum region, which affects motor function, cognitive, and decision-making processes [?]. Low dopamine levels may affect fundamental bodily processes, including dystonia, sleep disorders, depression, and cognitive changes [?].

The PD diagnosis is subjective and expert-dependent, following a visual assessment of motor symptoms during clinical procedures [?]. Such observational analysis is founded on standard scales based on the severity of motor and some non-motor symptoms. These diagnosis scales are subjective with coarse scales to stratify the disease, difficulting personalized treatment planning [?]. These reports indicate that the first PD diagnosis is made when already about 80% of the dopaminergic neurons are lost [?]. In consequence, late treatment reflects motor complications, the worsening of non-motor symptoms such as depression, and, cognitive decline, impacting the quality of the patient life-limiting the independence in essential tasks [?].

Today, neuroimaging analysis has emerged as a potential alternative for PD diagnosing, providing mechanisms to quantify volumetric affected brain regions and to characterize radiological findings associated with dopaminergic alterations, atrophy patterns, and cortical thickness [?]. On the one hand, T1-weighted Magnetic Resonance Imaging (MRI) enables the measurement of volumetric brain changes in critical brain areas such as the striatum, caudate nuclei, and putamen, facilitating the observational detection or exclusion of other forms of Parkinsonism, through manual morphometric measures such as the magnetic resonance Parkinson index (MRPI) [?]. Nevertheless, structural MRI images remain insufficient to detect small anomaly changes, especially in the subcortical region, making the visual inspection of these tiny brain regions an open challenge even for expert radiologists [?]. On the other hand, complementary SPECT studies (Single-photon emission computed tomography) have allowed to assess dopaminergic depletion in subcortical brain regions, which include the substantial nigra, before the onset of motor symptoms. Such SPECT modality employs an injection of a radioactive compound to trace dopaminergic pathways in the subcortical regions, measuring the dopaminergic neuron density improving the early diagnosis of the disease [**?**]. Although this modality is suggested for accurate diagnosis, there are some limitations regarding its use and availability in clinical settings [**?**].

This work introduces a self-supervised deep representation for recovering dopaminergic SPECT embedding descriptors, with the ability to complement structural information and discriminate between Parkinson's and control patients. This analysis is extended to Parkinsonian subjects on early symptoms onset (prodromal), and PD subjects without radiological findings (SWEDD). Following a T1-MRI to SPECT translation scheme the proposed approach is able to learn sequence correspondences and after that recover SPECT embedding descriptors from T1-MRI, with the promise to overcome the lack of SPECT studies in clinical routine. A preliminary version of this work was published in . A repository with the code of the proposed approach is available [here]. The specific contributions of this work are:

1. A self-supervised strategy to encode typical SPECT patterns from T1-MRI sources. Then, a PD vs control is adjusted to guarantee clean and well-supported labels. From such SPECT embedding, a discrimination analysis was made over Parkinsonian associations for prodromal and SWEDD patients as well for diagnosed PD (with a motor scale between 1 and 4).
2. A deep learning study with a total of three generators architectures were here validated to produce embedding representations that potentially encode complex relationships that represent PD

2 Related works

Imaging markers have been explored over diagnostic images, focusing principally on the extraction of morphological and textural descriptors. These descriptors were then subjected to statistical analysis and used to train supervised models to classify between abnormal and healthy observations [**?**], [**?**]. Such approximations, however, may be biased by the expert's criteria, which have been reported with low confidence in their ability to accurately bound abnormal patterns associated with parkinsonism. Conversely, alternative approaches have been developed to adjust deep representations from unsupervised tasks, evaluating their ability to reconstruct sequences from healthy reference cohorts [**?**]. For instance, a T1-MRI approach was dedicated to morphological coarse image characterization, but losing the identification of structural regions associated to dopaminergic damage which is crucial to define early PD signs and assessing the severity of the disease [**?**].

Regarding dopaminergic quantification, SPECT images (Single Photon Emission Computed Tomography) have been the recommended modality for measuring dopaminergic pathways and depletion in the substantia nigra, providing sufficient sensitivity to distinguish between PD patients along the stages [**?**]. From SPECT images have been designed and classified morphology, intensity, radial, and, gradient handcrafted features [**?**] [**?**]. Other approach, include motor clinical stratification, together with SPECT findings to predict the PD outcome, being a more sensitive score, rather than coarse the unified Parkinson's disease rating scale (UPDRS) scale. Nonetheless, due to the photon scatter and the collimator properties, SPECT sequences provide functional information rather than detailed anatomical information due to low spatial resolution, making PD characterization challenging without anatomical and structural reference [?], [?]. Even worse, SPECT is a specialized source with limited availability in clinical centers, which require radioactive substances. Besides, current computational techniques use strong supervised rules to update representations, which may be biased due to lower variability in training and the subjectivity on regions selection.

Anatomical-functional complementary information have been poorly explored, losing the natural codification of PD digital biomarkers, and a regulazer of false positive and negative PD regions. Regarding this correlative task, image-to-image translation and generative representations have been proposed to emulate asymmetry, and shape of the striatum, following a generative image transformer (GIT). This method, assesses the symmetric alterations rather than a diagnosis prediction. In a quantitative method, the striatal specific binding ratios (SBRs) commonly obtained across the SPECT modality, are derived from high-resolution MRI images like susceptibility-weighted imaging (SWI) [?]. In this case, the SWI and SPECT agreement is calculated using a regression task across volumetric patches surrounding the ROI. Such experimental approach remains restricted to routine scenarios because the availability of SPECT and high-resolution of MRI images.

3 Materials

Progressive dopamine neurotransmitter depletion in the substantial nigral region remains a hallmark in the Parkinson's disease (PD) diagnosis. PD exhibits multifactorial behavior evidenced by motor and non-motor symptoms. Nevertheless, someone with one or more of these symptoms does not necessarily have PD; in fact, these symptoms arise in other related parkinsonism in the early course of the disease [?]. The progression of PD in its early stages starts from neurodegeneration, where the typical motor aspects do not progress until a definitive clinical diagnosis is made. In this regard, the Movement Disorder Society (MDS) has proposed three main PD stages: preclinical (where the neurodegeneration has begun), prodromal (symptoms and signs are present but insufficient to make a definitive diagnosis), and clinical phase (diagnosis based on classical motor symptoms). Nevertheless, there exists a risk phase that involves some demographic features like gender (being men 1.5 times more affected than women [?]), age, genetic conditions, and, environmental exposure like pesticides and heavy metals [?]. In this phase, the first brain area to be affected is the substantia nigra (mainly involved in the control movement and coordination). In this sense, in the preclinical phase, subtle changes in daily autonomous activities begin to appear. The brain changes start in the serotonin system (brainstem raphe nuclei) mainly responsible for mood regulation, sleep patterns, and gastrointestinal regulation [?]. Interestingly, about 10 years before the formal clinical diagnosis (based on motor aspects), non-motor symptoms such as cognitive changes, attention issues, sleep disturbances, and sensory abnormalities arise in many patients at the onset of PD [?]. This phase is known as the prodromal stage and reveals more marked cognitive changes being evident on the olfactory bulb, locus coeruleus, and substantia nigra brain regions [?]. In this phase, these abnormalities could be detected on SPECT and

PET modalities where motor impairment is poorly perceptible being unfeasible in a formal clinical diagnostic. Even more interesting, when around 80% of dopaminergic neurons are damaged, common motor symptoms such as rigidity, tremor, bradykinesia, postural instability, and movement disorders appear and a format clinical diagnosis can be made following standard scales like Hoehn and Yahr scale [?]. In addition to that PD categorization, the Scan Without Evidence of Dopamine Deficit (SWEDD) group has gained the attention of the research community due to their PD diagnosis but their radiological findings look normal [?]. In this work, the prodromal and SWEDD groups are included and explained as follows:

3.1 SWEDD:

The Scan Without Evidence of Dopamine Deficit refers to the absence of image abnormalities on dopaminergic pathways in patients who are presumed to have PD [?]. There exists evidence that many patients on the SWEDD label have a variety of clinical conditions shared with PD condition. Due to the unusual nature of the clinical diagnosis, the SWEDD diagnosis and the related clinical aspects are still controversial in the research community. Some authors claim that the SWEDD classification is simply a product of misinterpretation of other neurodegenerative conditions [?]. Contrary, other works suggest SWEDD as a subtype of the PD group due to the positive levodopa response, clinical progression, or genetic evidence [?]. Some reports suggest that this group represents at least 20% of the PD population where the formal diagnosis was achieved on solely clinical features [?], [?]. SWEDD patients usually have an intact olfactory function, and normal gait, but, do not develop metabolic fingerprint key feature of PD, being suggested as an alternative diagnosis [?]. Nevertheless, a postmortem study of 21 SWEDD patients, showed that 16 subjects had nigral cell loss whereas 5 did not. In this sense, the scientific community suggests the positive DaT scan as the gold standard technique being more sensitive than the clinical examination for measuring the nigrostriatal damage [?].

3.2 Prodromal:

Taking into account that PD starts progressively, during early stages motor and non-motor symptoms could emerge being the non-motor aspects the first to arise [?]. Prodromal condition refers to the PD stage wherein early signs of damage in brain cells typically associated with PD are present, but the classic motor aspects is not. In other words, the prodromal cohort is the time before typical motor symptoms appear but neurodegeneration is present. It is worth to remark that the prodromal phase is not a preclinical PD' stage. Preclinical condition involve neurodegeneration but not evidence of motor symptoms). Prodromal stage present motor aspects but their remain insufficient to achieve a formal disease diagnosis [?]. Therefore, a classical clinical diagnosis (fully involve motor aspects) is not yet possible [?]. Due to the impossibility to perform a classical clinical assessment, the prodromal stage estimation is performed by the integration of diagnostic information like genetic findings, environment exposure, sex, and age. Nevertheless, the isolated diagnostic information remains insufficient and achieve a prodromal phase identification with 100% of certainty [?]. Therefore, an accurate identification of prodromal subjects is critical for providing early treatments that can reverse the disease's natural course and associated impairment. In this sense, the medical community has proposed a list of prodromal clinical motor, non-motor and neuroimaging markers to aid in the identification of this phase (see Figure ??).

4 Methods

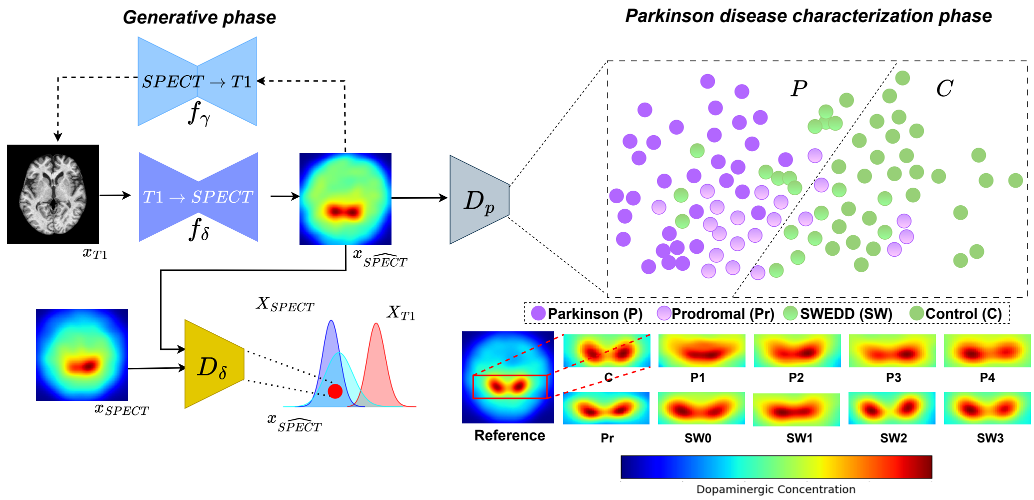

Fig. 1 Pipeline of the proposed method. In the generative phase, an image-to-image translation task is proposed to retrieve the more related dopaminergic pathways from T1-MRI input sequences through a CycleGAN-like architecture. The synthetic dopaminergic information adaption is measured by the discriminator net. In the second step, the dopaminergic descriptors are mapped to a CNN classifier to achieve the classification between Parkinson's and control populations. Prodromal and SWEDD subjects are mapped as well as additional test sets. In addition to the CNN backbone, an embedded feature representation could be seen as a classification alternative in the general binary classification task.

The pipeline of the proposed approach is illustrated in Figure 1. Firstly, the pre-processed T1-MRI and SPECT radiological samples are mapped to the CycleGan network following an unpaired scheme to retrieve dopaminergic descriptors from the T1-MRI representation. Hence, the decoder scheme in the f_δ architecture estimates the more related SPECT dopaminergic representation that serves as input to the CNN classifier. In this point, the embedding descriptors allow a PD and control discrimination populations. In addition, prodromal and SWEDD subjects are taken as additional test samples to measure the classifier capabilities with respect to these unseen and challenging populations.

4.1 Synthetic SPECT dopaminergic pathways from T1-MRI representations

The sensitive PD diagnosis is achieved through the dopaminergic pathways analysis over SPECT sequences. In this regard, we pretend to complement the anatomical information from typical T1-MRI observations with the most related functional dopaminergic representation in an unpaired T1 MRI-to-SPECT translation pre-text task. This dopaminergic estimation is approximated with a CycleGAN scheme composed of two generative encoder-decoder networks: δ and γ. Particularly, δ net approximates the dopaminergic representation (typically observed in SPECT sequences) but from anatomical T1-MRI inputs. Nevertheless, this isolated approximation may fall into meaningless clinical representations due to biased mapping. In this sense, the γ scheme performs a reverse mapping to recover the original anatomical T1-MRI representation from the synthetic SPECT representation. Hence, the integrity of the reconstruction samples is forced with a cycle consistency loss $L_{cyc}(\delta, \gamma)$ [**?**] following the expression:

$$\begin{aligned} L_{\mathrm{cyc}}(\mathbf{f}_{\boldsymbol{\delta}}, \mathbf{f}_{\boldsymbol{\gamma}}) =& \mathbb{E}_{\boldsymbol{x}_{\mathrm{T1}} \sim P_{\mathrm{data}}(\boldsymbol{x}_{\mathrm{T1}})} \left[\left\| \mathbf{f}_{\boldsymbol{\gamma}}(\mathbf{f}_{\boldsymbol{\delta}}(\boldsymbol{x}_{\mathrm{T1}})) - \boldsymbol{x}_{\mathrm{T1}} \right\|_1 \right] \\ &+ \mathbb{E}_{\boldsymbol{x}_{\mathrm{SPECT}} \sim P_{\mathrm{data}}(\boldsymbol{x}_{\mathrm{SPECT}})} \left[\left\| \mathbf{f}_{\boldsymbol{\delta}}(\mathbf{f}_{\boldsymbol{\gamma}}(\boldsymbol{x}_{\mathrm{SPECT}})) - \boldsymbol{x}_{\mathrm{SPECT}} \right\|_1 \right] \end{aligned} \tag{1}$$

In this context $\gamma(\delta(x_{T1})) \approx x_{T1}$, $\delta(\gamma(x_{SPECT})) \approx x_{SPECT}$ ensures that synthetic representations belong in data distribution X_{T1}, X_{SPECT}. In this sense, during the translation pre-text task the δ architecture (specialized to obtain SPECT from T1-MRI inputs) should learn paired dopaminergic patterns impacting the Parkinson's characterization. These dopaminergic features (spatially matched with T1-MRI dimensions) come from the δ decoder activations and are enriched using skip connections between the encoder and decoder of the δ generator scheme.

Additionally, a couple of *PatchGan* discriminatory architectures ensure the proper adaption to for both T1-MRI and SPECT radiological domains. In this sense, the dopaminergic pathways are retrieved from anatomical T1-MRI in an unpaired scheme. Figure 1 shows a *PatchGan* net (D_δ) specialized in the SPECT domain adaption. Hence, the dopaminergic descriptors from T1-MRI sequences are improved as:

$$L_{\mathbf{f}_{\boldsymbol{\delta}}} = L_{cyc} + L_{T1 \rightarrow SP\hat{E}CT} + L_{SPECT \rightarrow SP\hat{E}CT} \tag{2}$$

In this case, L_{cyc} is from equation 1. The $L_{T1 \rightarrow SP\hat{E}CT}$ optimizes the dopaminergic synthesis capability ($SP\hat{E}CT$) from unaligned $T1$ inputs. This loss function is defined as:

$$\begin{aligned} L_{\mathrm{T1} \rightarrow \mathrm{SP\hat{E}CT}} =& \mathbb{E}_{x_{\mathrm{SPECT}} \sim P_{\mathrm{data}}(\boldsymbol{x}_{\mathrm{SPECT}})} [\log D_\delta(x_{\mathrm{SPECT}})] \\ &+ \mathbb{E}_{x_{\mathrm{T1}} \sim P_{\mathrm{data}}(\boldsymbol{x}_{\mathrm{T1}})} [\log(1 - D_\delta(\mathbf{f}_{\boldsymbol{\delta}}(x_{\mathrm{T1}})))] \end{aligned} \tag{3}$$

The $L_{SPECT \to SP\hat{E}CT}$ is an identity loss that measure the conservation of dopaminergic patterns being defined as: $L_{SPECT \to SP\hat{E}CT} = [\|\mathbf{f}_{\boldsymbol{\delta}}(\boldsymbol{x}_{\text{SPECT}}) - \boldsymbol{x}_{\text{SPECT}}\|_1]$. Therefore, the full loss function could be summarized as:

$$L_{total}(\mathbf{f}_{\boldsymbol{\delta}}, \mathbf{f}_{\boldsymbol{\gamma}}, D_\delta, D_\gamma) = L_{\mathbf{f}_{\boldsymbol{\delta}}} + L_{\mathbf{f}_{\boldsymbol{\gamma}}} + L_{cyc}(\mathbf{f}_{\boldsymbol{\delta}}, \mathbf{f}_{\boldsymbol{\gamma}}) \tag{4}$$

4.2 Automatic Parkinson's disease discrimination from synthetic dopaminergic pathways

The coherence of the dopaminergic descriptors from T1-MRI anatomical inputs is validated by an adjusted CNN classifier (D_p in Figure 1) over SPECT radiological sequences. In this regard, given an anatomical T1-MRI sequence, the corresponding dopaminergic descriptor is retrieved from the $\delta(x_{T1})$'s decoder network and then, fed as input to the CNN classifier in a second phase achieving a PD/control discrimination task. Additionally, from such D_p, a latent dopaminergic representation is retrieved with the main learned patterns between the anatomical and dopaminergic information, allowing the binary discrimination task. From such embedded representation, a 3D projection is made through the PCA algorithm. With this in mind, is worth noticing that a neighboring analysis is possible to measure the dopaminergic pathways associated with the PD.

5 Experimental setup

5.1 Data

The proposed approach was validated over a cohort of Parkinson's patients, controls, prodromal, and SWEDD subjects obtained from the publicly available PPMI database [**?**]. PPMI is one the most relevant landmark effort with an international collaboration to analyze the PD progression from biological markers that include several neuroimaging modalities, genetics, and biospecimen information. From this database, we select a cohort with patients with available T1-MRI and SPECT sequences for the generative phase, and only SPECT sequences for the PD/Control classification task. The Figure xxx shows the inclusion criteria taken into account regarding each computational task. For the generative task, we consider a total of 75 control subjects and 75 PD-diagnosed patients stratified regarding the Hoehn and Yahr scale for the SPECT modality. Regarding the T1-MRI sequences, a total of 80 PD patients and 73 control subjects were taken into account. From such selection criteria, the PD cohort reports an average delay larger than 255 days between T1-MRI and SPECT image acquisitions, and 25 days for control subjects. For this reason, the T1-MRI and the related SPECT sequences are mainly unpaired. Table 1 shows the demographic information of the considered sample of subjects.

Regarding the classification task, we collect a total of 343 PD patients and 337 control subjects hierarchically ranked regarding the Hoehn and Yahr scale. Table 2 shows the demographic data. Then, the training phase is composed by 80% of the PD and control subjects for the generative and classification tasks, and the remaining 20% were reserved to test the performance of both tasks. In this proposed approach, all the

Domain	Variable	Train	Test
	Parkinson/Control	60 / 60	15 / 15
Spect	Age (mean±std)	64.01±9.89 / 60.24±11.83	66.07±8.91/ 61.47±10.84
2-4	H&Y scale (1/2/3/4)	8 / 49 / 1 / 2	2 / 11 / 1 / 1
	Parkinson/Control	64 / 58	16 / 15
T1-MRI	Age (mean±std)	64.02±9.89 / 60.1±11.98	66.07±8.91 / 61.47±10.84
2-4	H&Y scale (1/2/3/4/Unk)	8 / 47 / 1 / 2 / 6	2 / 10 / 1 / 1 / 2

Table 1 Demographic information and the related H&Y scale of the considered PD subjects regarding the T1-SPECT translation task.

neuroimage sequences were acquired from different scan manufacturers (SIEMENS, Marconi Medical Systems, PICKER, GE MEDICAL SYSTEMS)

Variable	Train	Test
Parkinson/Control	273 / 273	70 / 64
Age (mean ± std)	64.02 ± 9.89 / 62.22 ± 11.6	66.07 ± 8.91 / 63.48 ± 11.02
H&Y scale (1/2/3/4)	70 / 190 / 9 / 4	17 / 48 / 2 / 3

Table 2 Demographic information and the related H&Y scale of the considered PD subjects regarding the SPECT classification task.

The collected T1-MRI and SPECT sequences were spatially normalized considering isometric resampling of data with a voxel spacing of $1cm^3$. Particularly, for the T1-MRI data the skull was removed using the synthstrip utility of the freesurfer software. To ensure that same tissue type has the similar intensities values along the study, a N4 bias field correction was applied using the SimpleITK package. Subsequently, each T1-MRI was registered with a rigid method to the MNI-152 space implemented with ANTsPy [**?**]. On the other hand, after the isometric resampling step, an isotropic Gaussian filter with 18-mm full-width-at-half-maximum was applied to ensure a low spatial resolution. Once all the preprocessing steps were applied, the preprocessed images were interpolated to $182 \times 218 \times 182$-mm and then converted to png files to fit the generative and CNN networks. Once the preprocessing steps were applied, this proposal considered slices in axial view. As reported in the medical field, prodromal and SWEDD groups present a challenging nature being highly misclassified. In this sense, the proposed approach was fine-tunned to characterize between control and PD subjects, were prodromal and SWEDD groups were used as additional test sets in order to measure the model's sensitivity. Is worth to notice that for the prodromal set, only the T1-MRI sequences are available collecting a total of 46 subjects with T1-MRI images. Regarding the SWEDD cohort, a total of 79 subjects were taken into account were 39 have both image studies captured.

5.2 Model setup

The proposed approach include *Unet* blocks for the couple of generators ($\mathbf{f}_\delta$, $\mathbf{f}_\gamma$) that performs the $T1 \rightarrow SPECT$ and $SPECT \rightarrow T1$ mapping, respectively. The *Unet* architecture contains enconder and decoder blocks ensuring the feature preservation along the training process. Particularly, each *Unet* net has 16 layers with 54.4 million parameters. The $\mathbf{D}_{\boldsymbol{\delta}}$ is a *PatchGan* network (reporting around 2.7 million parameters) useful for differentiate the original from synthetic distributions in a patch pixels average analysis [**?**]. *PatchGan* scheme includes a set of convolutional layers with a number of kernels: 64-128-256-512, with weights initialized from a Gaussian distribution with values of 0 and 0.02 for mean and standard deviation, respectively. Regarding the PD vs Control differentiation from SPECT radiological sequences, this approach includes a CNN scheme. In such a case, a *MobileNet* architecture was formerly trained SPECT training set. Throughout the training process, the architecture was fine-tuned using the Adam optimizer with a learning rate of 2×10^{-4} over 50 epochs. The image input is 256×256 with around 2250 and 1377 relevant frames for SPECT and T1-MRI domain in the translation scheme, respectively. On the other hand, for the classification task, we take into account around 4965 SPECT relevant frames. Once the CNN classifier scheme was adjusted, a fine-tuning step was performed with the converted T1-MRI training set employing the trained XXX scheme. In this point, the PD/Control discrimination is performed by majority voting over all the slices that compose the patient's case volume. Lastly, from such CNN classifier, an embedding descriptor was retrieved from the softmax previous layer with a dimensional size of xxx. In this regard, a simple XXXX method was trained to classify among PD/Control classes in such topological space.

6 Evaluation and Results

6.1 Parkinson's classification

To validate the coherence in the dopaminergic mapping from standard T1-MRI sequences, an ablation study was carried out using different kind of convolutional backbones in the control vs. Parkinson classification task. The table 3 shows the classification performance revealing superior results from the VGG16 using the precision and harmonic mean as score criteria. Nevertheless, it is to worth to notice that the Mobilenet architecture evidences similar competitive results reporting a lower false negative predictions improving the recall metric by 1% regarding Vgg16 scheme. These promising results of the VGG16 could be due to this scheme employs classical convolutional stragies that benefit the primary properties like edges and contours. On the other hand, lighther strategies like MobileNet architecture, uses depthwise and pointwise separable convolutions reducing the amount of parameters and being useful in the preservation of key image features. Interestingly, more recently schemes like ConvNeXt, inspired by the design of Vision Transformers, also include depthwise convolutions, inverted bottleneck design and large kernel sizes. Therefore, this architecture improves significantly the computational efficiency reducing the amount of hyperparameters to be tuned but being prone to overfitting.

Figure 2 shows the top two models performance (VGG16 and MobileNet) in the Parkinson vs Control classification task. The boxplots show the Parkinson's probability distribution (y-axis) over all the SPECT slices (control and Parkinson) on the x-axis. In this sense, is to be expected that control slices have lower Parkinson's probabilities while high probability values for Parkinson's slices. In this regard, both architectures have similar performance being slighter confident in the Parkison's detection (reduce

Net	Acc	Pre	Rec	F1-score
Vgg16	**96%**	**97%**	95%	**96%**
ConvNeXt	82%	76%	92%	83%
MobileNet	96%	96%	**96%**	96%
ResNet50	56%	76%	56%	44%

Table 3 Abaltion study over test set in the PD/control classification task.

amount of Parkinson's cases labeled as control) the MobileNet architecture explaining the 1% higher recall result regarding the VGG16 scheme.

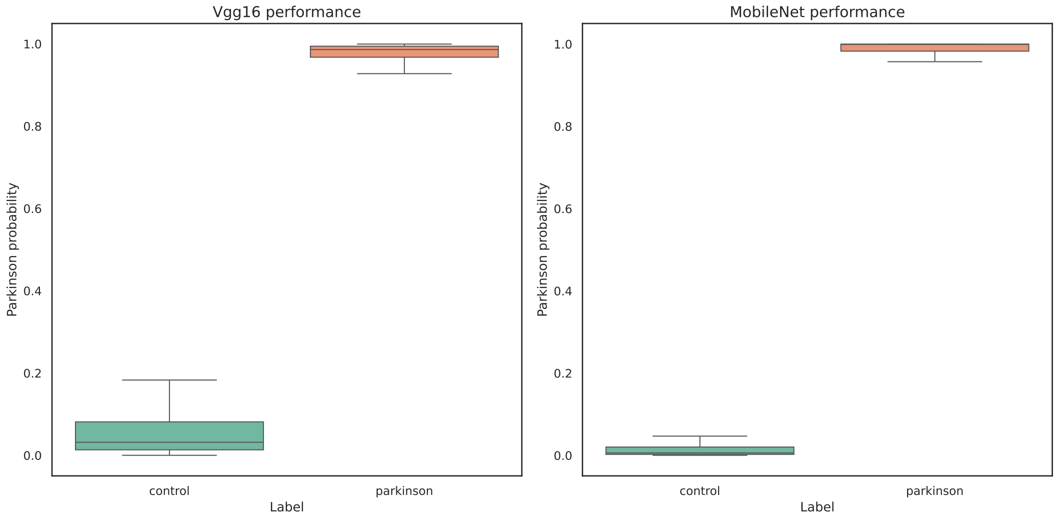

Fig. 2 Boxplot of the top two best CNN schemes retrieving the Parkinson's related probability.

From such VGG16 fine-tunned over SPECT domain, in a second step the discrimination capability of the proposed approach following the translation is measured. In such a case, we considered two alternatives to achieve the Parkinson's discrimination: mapped the synthetic dopaminergic images (closer SPECT estimation) to the D_ρ Parkinson's classifier (VGG16) and the retrieved embedding space from such classifier using a xxxx strategy. Table 4 shows the classification performance achieved over the different radiological sequences, the related dopaminergic estimation employing different state-of-the-art image translation strategies, and the reference SPECT performance.

As expected, the T1-MRI classification achieves the lowest results. This could be due to the slight morphological differences present in the early stages of the disease that can make difficult the discrimination task. Interestingly, once the T1-MRI sequences are feeded to the translation phase, the enhanced dopaminergic descriptor version outperforms by xxx these anatomical representations. This results suggest a coherent dopaminergic mapping into the anatomical information highly related with the typical

Type of data	Accuracy	Precision	Recall	F1-score
T1-MRI	61	61	61	61
Unet descriptor	48	47	48	46
Pix2Pix descriptor	52	51	51	51
CycleGan descriptor	58	58	58	58
SPECT	**96%**	**97%**	**95%**	**96%**

Table 4 Metrics performance from T1-MRI, the dopaminergic descriptor retrieved by different state-of-the-art schemes, and referent SPECT sequences. *All values represent percentages. FALTA SUPERAR NUMEROS

Parkinson's properties. Nevertheless, the achieved results are significant lower than the the reference SPECT, suggesting that additional tuned steps should be done in the translation phase to encode properly discriminative dopaminergic information.

No fold	Acc	Pre	Rec	F1-score
1	**0,96**	**0.97**	**0.95**	**0.96**
2	0,96	0,96	0,96	0,96
3	0,96	0,97	0,94	0,96
4	0,95	0,98	0,92	0,95
5	0,95	0,96	0,95	0,95
avg	0,96	0,97±0,01	0,94±0,01	0,96

Table 5 Five cross-validation performance for the original SPECT classifier. The top row results correspond with the initial train/test split.

6.2 Prodromal and SWEDD populations classification

6.3 Validation of image synthesis

Type of data	Accuracy	Precision	Recall	F1-score
T1-MRI	xxx	xxx	xxx	xxx
Dopaminergic descriptor	xxx	xxx	xxx	xxx
SPECT	xxx	xxx	xxx	xxx

Table 6 **Embedding classification performance from the CNN classifier. *All values represent percentages**

Metric		CycleGan	Pix2Pix	Unet
PSNR	Control	21.83±0.37	**25.58±3.1**	24.26±3.61
	PD	19.3±2.19	**23.46±2.89**	23.06±2.65
SSIM	Control	63.03±0.04	**73±0.1**	67.59±0.15
	PD	43.83±0.1	**67±0.1**	63.65±0.09

Table 7 Image translation quality employing different kind of translation schemes regarding the type of population

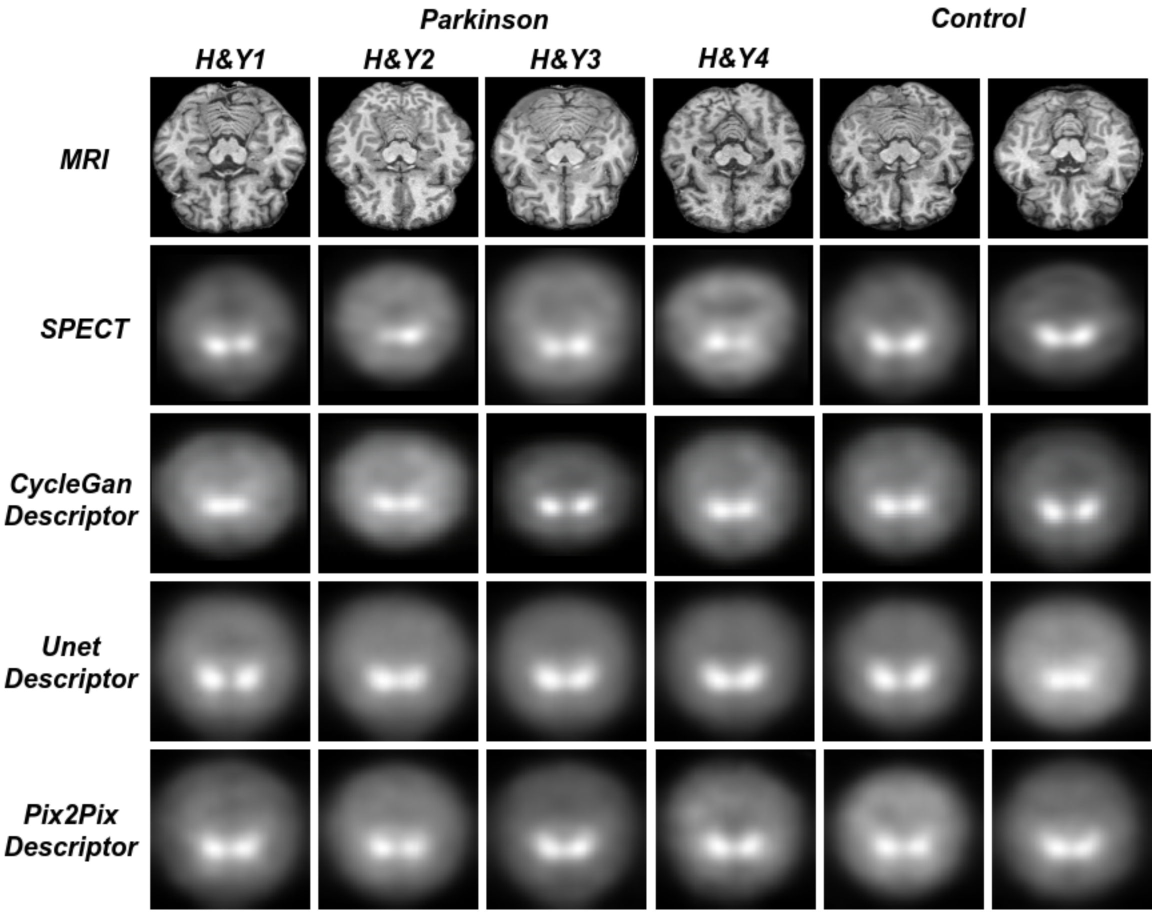

Fig. 3 Some visual results

7 Discussion

8 Conclusion

Supplementary information.

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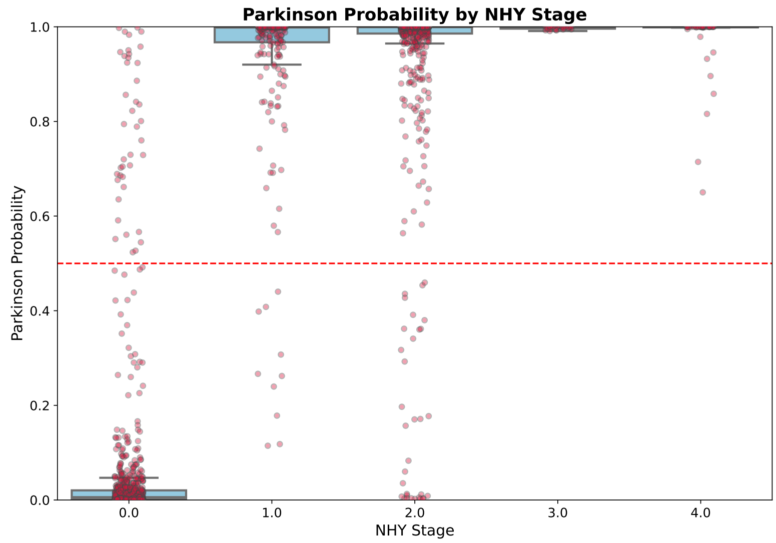

Fig. 4 Parkinson prediction over control and Parkinson populations regarding the HY stage.

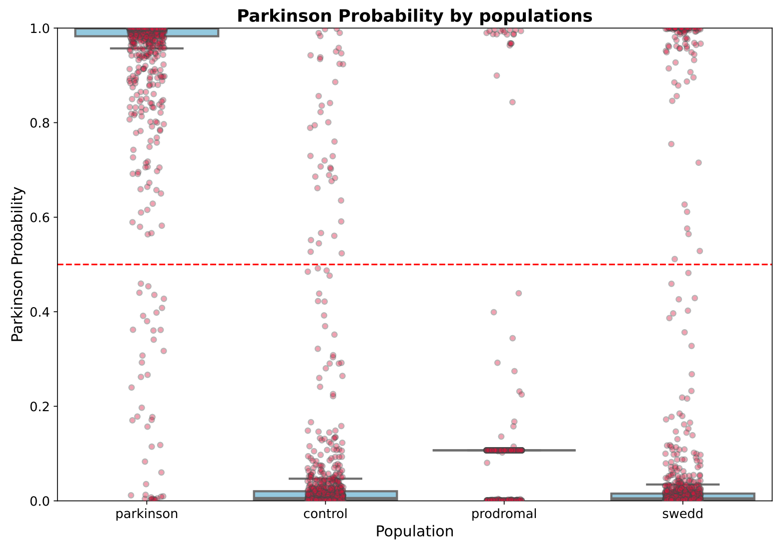

Fig. 5 Parkinson prediction over all the population samples that were taken into account in this study.

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9 Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.